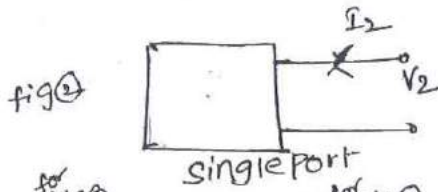
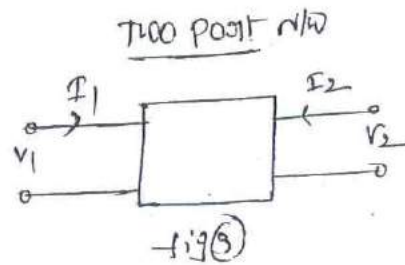
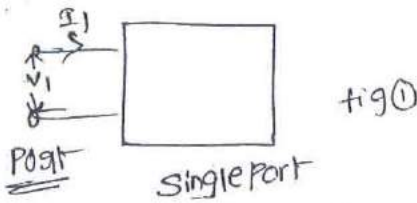


Two port N/w



Ex: for fig 1
Motor

i/p → Electrical
o/p → Mechanical

Ex: for fig 2

Generator

i/p → Mech
o/p → Elec

Ex: 1st pair of terminals = i/p Port
2nd " " = o/p Port

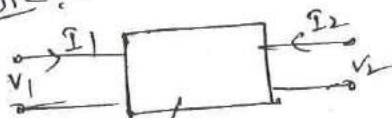
* A two port n/w always assumption is current enters in to the N/w from two ports.

Ex: T/F

- * → Pair of terminals at which signal may enter (or) leave from the N/w is called as "port".
- * when the N/w is having only one pair of terminals then the N/w is called as "single port N/w" (as shown in fig 1 & 3).
- * when the N/w is having two pair of terminals then the N/w is called as "Two Port N/w".

Ex: T/F → Two port N/w

Note: -



Complex N/w

↓
Combination of series parameters
parallel electronic devices
T/F

Classification of parameters: -

1. Z-parameters :- (Open circuit parameters)
2. Y-parameters :- (Short circuit parameters)
3. h-parameters :- (Hybrid parameters)
4. g-parameters :- (Inverse hybrid parameters)
5. ABCD parameters :- (Transmission line parameters)
6. A'B'C'D' (or) abcd parameters :- (Inverse Transmission line)

Z-parameters :-

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \rightarrow (1)$$

$$V_2 = Z_{21}I_1 + Z_{22}I_2 \rightarrow (2)$$

V_1 & V_2 = dependent variables
 I_1 & I_2 = independent variables

x_1 out of all $\begin{pmatrix} I_1 \\ I_2 \end{pmatrix}$ are sources

units

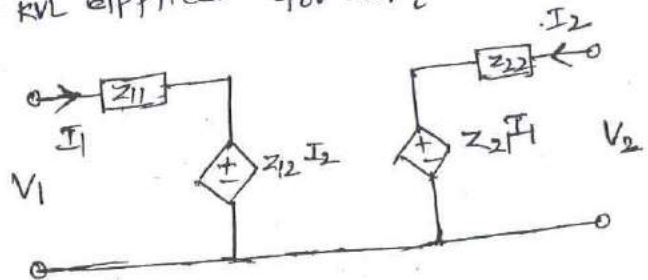
$$(Z) \quad Z_{11} = \left. \frac{V_1}{I_1} \right|_{I_2=0} \quad \left. \begin{array}{l} \text{Port 2 is} \\ \text{open ckt} \end{array} \right\}$$

$$(Z) \quad Z_{21} = \left. \frac{V_2}{I_1} \right|_{I_2=0}$$

$$(Z) \quad Z_{12} = \left. \frac{V_1}{I_2} \right|_{I_1=0} \quad \left. \begin{array}{l} \text{Port 1 is} \\ \text{open ckt} \end{array} \right\}$$

$$(Z) \quad Z_{22} = \left. \frac{V_2}{I_2} \right|_{I_1=0}$$

* eqn (1) & (2) are voltage eqns. so KVL applied.
 * KVL applied for series ckt.



Z_{11} = open ckt input impedance

(or) Driving point input impedance

Z_{21} = Forward Transfer Impedance

Z_{12} = Reverse Transfer Impedance

(since $V_1 = 1$ IP
 $I_2 = 0$ IP)

Z_{22} = open ckt output impedance (or)

Driving point output impedance.

Y-parameters :-

$$I_1 = Y_{11}V_1 + Y_{12}V_2 \rightarrow (1)$$

$$I_2 = Y_{21}V_1 + Y_{22}V_2 \rightarrow (2)$$

I_1 & I_2 → dependent var

V_1 & V_2 → independent var

V_1 & V_2 → sources since independence.

units

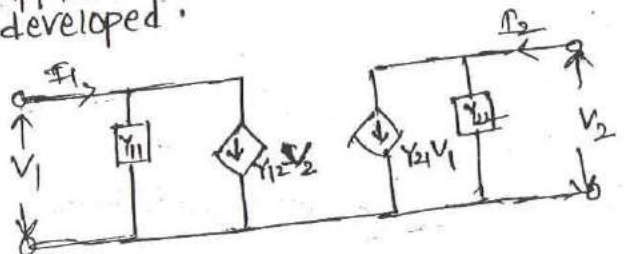
$$(Y) \quad Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} \quad \left. \begin{array}{l} \text{Port 2 is} \\ \text{short ckt} \end{array} \right\}$$

$$(Y) \quad Y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0}$$

$$(Y) \quad Y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0} \quad \left. \begin{array}{l} \text{Port 1 is} \\ \text{short ckt} \end{array} \right\}$$

$$(Y) \quad Y_{22} = \left. \frac{I_2}{V_2} \right|_{V_1=0}$$

* eqn (1) & (2) are current eqns. so KCL is applied. for that parallel ckt is developed.



Y_{11} = short ckt input admittance
 (or) Driving point input admittance

Y_{21} = forward transfer admittance

Y_{12} = Reverse "

Y_{22} = short ckt output admittance
 (or) Driving point output admittance

h-parameters :-

$$V_1 = h_{11} I_1 + h_{12} V_2 \rightarrow (1)$$

$$I_2 = h_{21} I_1 + h_{22} V_2 \rightarrow (2)$$

$\begin{pmatrix} V_1 \\ I_2 \end{pmatrix} \rightarrow \text{dependent}$

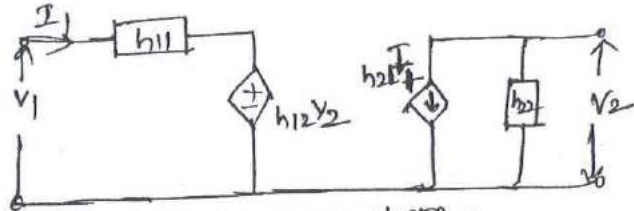
$\begin{pmatrix} I_1 \\ V_2 \end{pmatrix} \rightarrow \text{independent}$

$$h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0}$$

$$h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0}$$

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0}$$

$$h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0}$$



Relation b/w h & other parameters :-

$$1) \quad h_{11} = \left. \frac{V_1}{I_1} \right|_{V_2=0} \quad (\text{short circuit}) \quad (h_{11} \neq Z_{11}, h_{11} = \frac{1}{Y_{11}})$$

$$\hookrightarrow h_{21} = \left. \frac{I_2}{I_1} \right|_{V_2=0} \quad (\text{short circuit}) \quad (\text{by making } V_2=0 \text{ we get two Y-parameters, } Y_{11} \text{ \& } Y_{21})$$

$$h_{12} = \left. \frac{V_1}{V_2} \right|_{I_1=0} \quad (\text{open circuit})$$

$$h_{12} = \frac{V_1}{I_2} \cdot \frac{I_2}{V_2} = \frac{Z_{12}}{Z_{22}}$$

$$h_{21} = \frac{I_2}{I_1}$$

$$= \frac{I_2/V_1}{I_1/V_1} = \frac{Y_{22}}{Y_{11}}$$

$$2) \quad h_{22} = \left. \frac{I_2}{V_2} \right|_{I_1=0} \text{ o.c.}$$

$$= \frac{I_2}{V_2} = \frac{1}{Z_{22}}$$

g-parameters :-

$$I_1 = g_{11} V_1 + g_{12} I_2 \rightarrow (1)$$

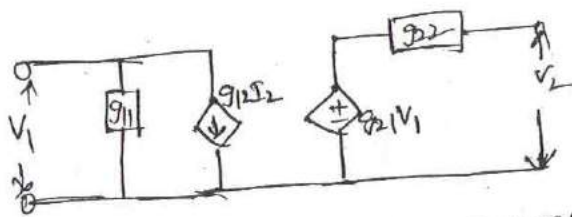
$$V_2 = g_{21} V_1 + g_{22} I_2 \rightarrow (2)$$

$$g_{11} = \left. \frac{I_1}{V_1} \right|_{I_2=0} \quad \frac{\text{units}}{V}$$

$$g_{12} = \left. \frac{I_1}{I_2} \right|_{V_1=0} \quad \text{Amps}$$

$$g_{21} = \left. \frac{V_2}{V_1} \right|_{I_2=0} \quad \text{Amps}$$

$$g_{22} = \left. \frac{V_2}{I_2} \right|_{V_1=0} \quad \Omega$$



Relation b/w g & other parameters :-

$$1) \quad g_{11} = \left. \frac{I_1}{V_1} \right|_{I_2=0} \text{ o.c.}$$

$$g_{11} = \frac{1}{Z_{11}}$$

$$2) \quad g_{12} = \left. \frac{I_1}{I_2} \right|_{V_1=0} \text{ s.c.}$$

$$3) \quad g_{21} = \left. \frac{V_2}{V_1} \right|_{I_2=0} \text{ o.c.}$$

$$= \frac{V_2}{I_1} \cdot \frac{I_1}{V_1} = \frac{Z_{21}}{Z_{11}}$$

$$= \frac{I_1/V_2}{I_2/V_2} = \frac{Y_{12}}{Y_{22}}$$

$$4) \quad g_{22} = \left. \frac{V_2}{I_2} \right|_{V_1=0} \text{ s.c.}$$

$$g_{22} = \frac{1}{Y_{22}}$$

$$g_{12} = \frac{Y_{12}}{Y_{22}}$$

ABCD Parameters :-

$$V_1 = AV_2 - B I_2$$

$$I_1 = CV_2 - D I_2$$

$$A = \left. \frac{V_2}{V_1} \right|_{I_2=0} \rightarrow \frac{\text{units}}{\text{Nounits}}$$

$$B = \left. -\frac{V_1}{I_2} \right|_{V_2=0} \rightarrow \Omega$$

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0} \rightarrow \text{u}$$

$$D = \left. -\frac{I_1}{I_2} \right|_{V_2=0} \rightarrow \text{Nounits}$$

Note:-

By using eq (1) & (2) it is not possible to develop eqn. ckt, since both eqns are developed w.r.to A/P.

A'B'C'D' or abcd parameters :-

$$V_2 = A' V_1 - B' I_1 \rightarrow (1)$$

$$I_2 = C' V_1 - D' I_1 \rightarrow (2)$$

$$A' = \left. \frac{V_2}{V_1} \right|_{I_1=0} \rightarrow \frac{\text{units}}{\text{Nounits}}$$

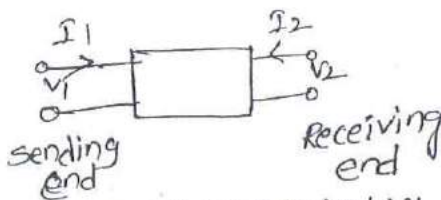
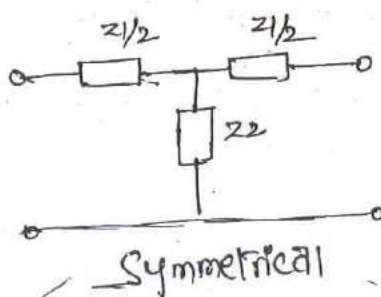
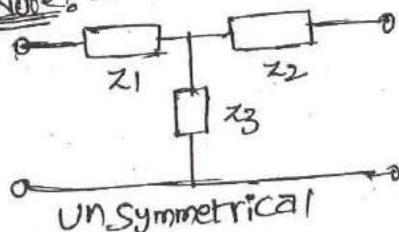
$$B' = \left. -\frac{V_2}{I_1} \right|_{V_1=0} \rightarrow \Omega$$

$$C' = \left. \frac{I_2}{V_1} \right|_{I_1=0} \rightarrow \text{u}$$

$$D' = \left. -\frac{I_2}{I_1} \right|_{V_1=0} \rightarrow \text{Nounits}$$

~~Recall~~

Note:-



* T/A represented by ABCD parameters

Relation b/w ABCD & other parameters :-

$$A = \left. \frac{V_2}{V_1} \right|_{I_2=0} \text{ Open ckt}$$

$$= \frac{V_2}{I_1} \cdot \frac{I_1}{V_1} = \frac{Z_{21}}{Z_{11}} \dots$$

$$B = \left. -\frac{V_1}{I_2} \right|_{V_2=0} \text{ S.c}$$

$$= -\frac{1}{Y_{21}}$$

$$C = \left. \frac{I_1}{V_2} \right|_{I_2=0} \text{ O.c} = \frac{1}{Z_{21}} \dots$$

$$D = \left. -\frac{I_1}{I_2} \right|_{V_2=0} \text{ S.c}$$

$$= -\frac{I_1/V_1}{I_2/V_2} = -\frac{Y_{11}}{Y_{21}} \dots$$

$\Rightarrow \dots$

Relation b/w A'B'C'D' & other parameters :-

$$A' = \left. \frac{V_2}{V_1} \right|_{I_1=0} \text{ O.c}$$

$$\frac{V_2}{I_2} \cdot \frac{I_2}{I_1} = \frac{Z_{22}}{Z_{12}}$$

$$B' = \left. -\frac{V_2}{I_1} \right|_{V_1=0} \text{ S.c}$$

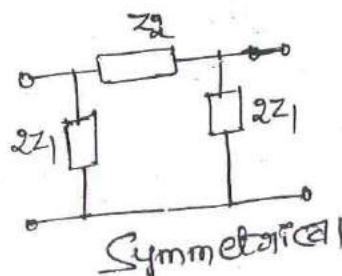
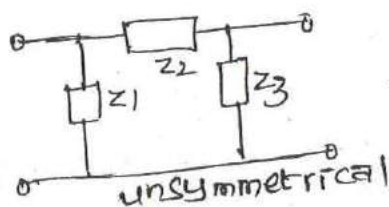
$$B' = -Z_{21}$$

$$C' = \left. \frac{I_2}{V_1} \right|_{I_1=0} \text{ O.c}$$

$$C' = \frac{1}{Z_{12}}$$

$$D' = \left. -\frac{I_2}{I_1} \right|_{V_1=0} \text{ S.c}$$

$$D' = -\frac{I_2/V_2}{I_1/V_2} = -\frac{Y_{22}}{Y_{12}}$$



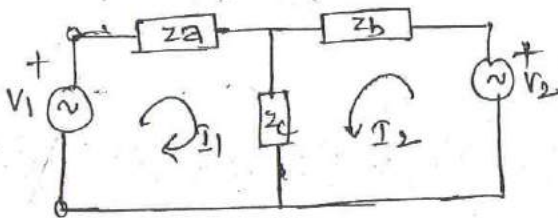
Note :-

Symmetrical	Reciprocal
$Z_{11} = Z_{22}$	$Z_{12} = Z_{21}$
$Y_{11} = Y_{22}$	$Y_{12} = Y_{21}$
$h_{11}h_{22} - h_{12}h_{21} = 1$	$h_{12} = -h_{21}$
$g_{11}g_{22} - g_{12}g_{21} = 1$	$g_{12} = -g_{21}$
$A = D$	$AD - BC = 1$
$A' = D'$	$A'D' - B'C' = 1$

Symmetrical N/w :-

Def :- Whenever the N/w looks the same from both the ports then it is said to be "symmetrical" - otherwise it is unsymmetrical.

T-N/w :-



$$\therefore Z_{11} = Z_a + Z_c$$

$$Z_{12} = Z_{21} = Z_c$$

$$Z_{22} = Z_b + Z_c$$

corresponds to loop ① & loop ②:

$$V_1 = (Z_a + Z_b) I_1 + Z_c I_2$$

$$V_2 = Z_c I_1 + (Z_b + Z_c) I_2$$

compare this with standard eqns

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

$$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} Z_a + Z_c & Z_c \\ Z_c & Z_b + Z_c \end{bmatrix}$$

That means by knowing the circuit constants we can find the Z-parameters

NOW find Z_a, Z_b, Z_c from ② parameters :-

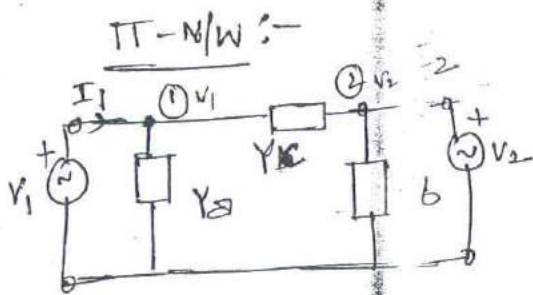
$$Z_{12} = Z_{21} = Z_c$$

$$Z_a + Z_c = Z_{11}$$

$$\Rightarrow Z_a = Z_{11} - Z_{12}$$

$$Z_b + Z_c = Z_{22}$$

$$\Rightarrow Z_b = Z_{22} - Z_{21}$$



$$\Rightarrow Y_{11} = Y_a + Y_c$$

$$Y_{12} = Y_c - Y_c = 0$$

$$Y_{22} = Y_b + Y_c$$

Note: $I_1 = Y_a V_1 + (Y_1 - Y_2) V_c$

$$I_1 = (Y_a + Y_c) V_1 - Y_c V_2$$

$$I_2 = (Y_b + Y_c) V_2 - Y_c V_1$$

By comparing this with standard form

$$I_1 = Y_{11} V_1 + Y_{12} V_2$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2$$

$$\therefore \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} = \begin{bmatrix} Y_a + Y_c & -Y_c \\ -Y_c & Y_b + Y_c \end{bmatrix}$$

Now find Y_a, Y_b, Y_c from Y-parameters :-

$$Y_{12} = Y_{21} = -Y_c$$

$$Y_{11} = Y_a + Y_c$$

$$Y_a = -Y_{12} - Y_c$$

$$Y_{22} = Y_b + Y_c$$

$$\Rightarrow Y_b = -Y_{22} - Y_c$$

Lattice N/W :-

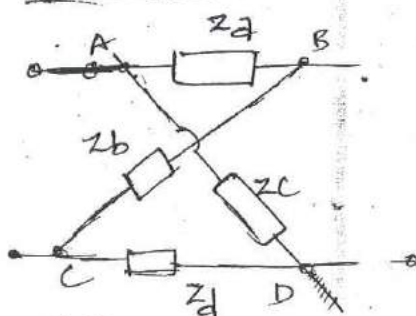
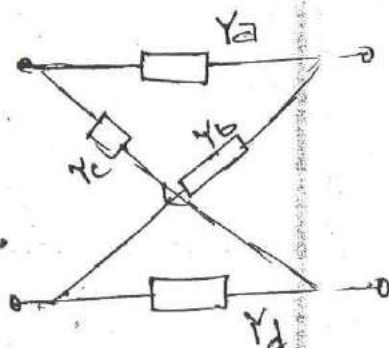


fig 1



$Z_a = Z_b, Z_c = Z_d$ } Symmetrical condition for lattice N/W

$$Z_{11} = Z_{22} = \frac{Z_b + Z_a}{2}$$

$$Z_{12} = Z_{21} = \frac{Z_b - Z_a}{2}$$

Note: Z_a & Z_b from Z_{11} & Z_{12}

$$Z_b = Z_{11} + Z_{12}$$

$$Z_a = Z_{11} - Z_{12}$$

$Y_a = Y_b, Y_c = Y_d$ } Symmetrical condn

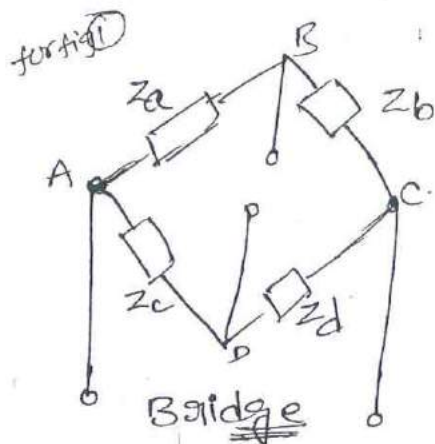
$$Y_{11} = Y_{22} = \frac{Y_b + Y_a}{2}$$

$$Y_{12} = Y_{21} = \frac{Y_b - Y_a}{2}$$

Note: Y_a & Y_b from Y_{11} & Y_{12}

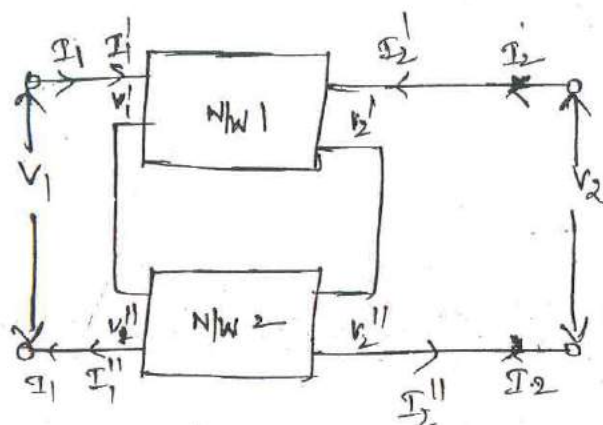
$$Y_b = Y_{11} + Y_{12}$$

$$Y_a = Y_{11} - Y_{12}$$



* Lattice N/w & Bridge N/w.
Both are same.

series connection of two two port N/w :-



for N/w 1 $\rightarrow$
for N/w 2 $\rightarrow$

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix}$$

$$\begin{aligned} V_1 &= V_1' + V_1'' \\ I_1 &= I_1' = I_1'' \end{aligned}$$

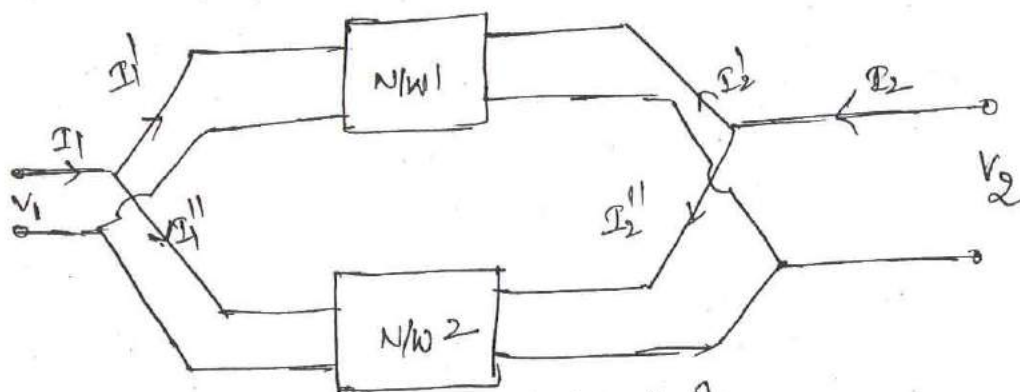
for port ①

$$\begin{aligned} V_2 &= V_2' + V_2'' \\ I_2 &= I_2' = I_2'' \end{aligned}$$

for port ②

$$\begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} = \begin{bmatrix} z_{11}' & z_{12}' \\ z_{21}' & z_{22}' \end{bmatrix} + \begin{bmatrix} z_{11}'' & z_{12}'' \\ z_{21}'' & z_{22}'' \end{bmatrix}$$

parallel connection of two two port N/w :-



for N/w 1 $\rightarrow$ $\begin{bmatrix} y_{11}' & y_{12}' \\ y_{21}' & y_{22}' \end{bmatrix}$

for N/w 2 $\rightarrow$ $\begin{bmatrix} y_{11}'' & y_{12}'' \\ y_{21}'' & y_{22}'' \end{bmatrix}$

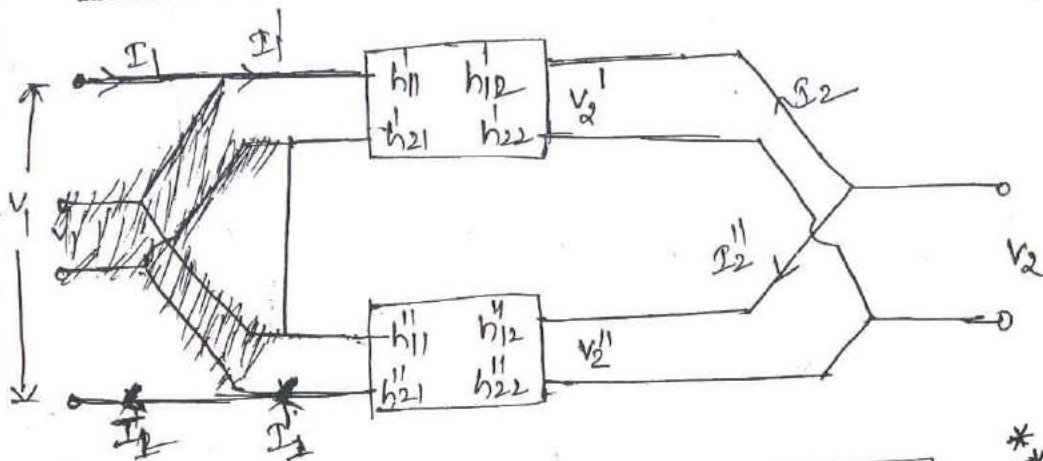
At port ① :- $V_1 = V_1' = V_1''$; $I_1 = I_1' + I_1''$

At port ② :- $V_2 = V_2' = V_2''$; $I_2 = I_2' + I_2''$

equiv γ parameters

$$\begin{bmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{bmatrix} = \begin{bmatrix} \gamma_{11}^I & \gamma_{12}^I \\ \gamma_{21}^I & \gamma_{22}^I \end{bmatrix} + \begin{bmatrix} \gamma_{11}^{II} & \gamma_{12}^{II} \\ \gamma_{21}^{II} & \gamma_{22}^{II} \end{bmatrix}$$

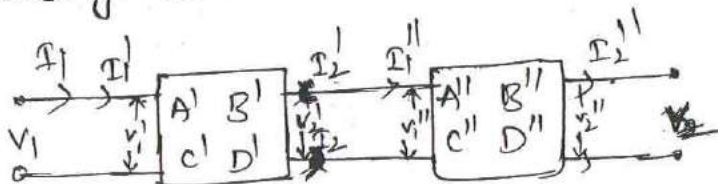
parallel connection of two 2-port N/W.S.:-



* i/p Port $\rightarrow$ series
* o/p Port $\rightarrow$ parallel

$$\begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} h_{11}^I & h_{12}^I \\ h_{21}^I & h_{22}^I \end{bmatrix} + \begin{bmatrix} h_{11}^{II} & h_{12}^{II} \\ h_{21}^{II} & h_{22}^{II} \end{bmatrix} //$$

cascading connection of two twoport N/W.S.:-



$$V_1 = V_1' \\ I_1 = I_1'$$

$$V_2' = V_1'' \\ I_2' = I_1''$$

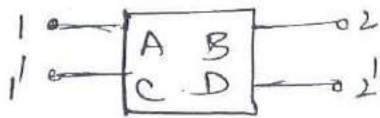
$$V_2 = V_2'' \\ I_2 = I_2''$$

W.K.T When ever Two gains are cascaded

$$\begin{matrix} \boxed{G_1} & \boxed{G_2} \\ \downarrow & \\ \boxed{G_1 G_2} \end{matrix}$$

$$\therefore \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix} //$$

[open circuit & short circuit]



$$\underline{\text{Sol:}} \quad (Z_{eq})_1 = \frac{V_1}{I_1} = \frac{AV_2 - B I_2}{CV_2 - D I_2}$$

$$(Z_{\text{eq}})_{\text{O.C}} = \frac{V_1}{I_1} \Big|_{I_2=0} = \frac{AV_2}{CV_2} = \frac{A}{C} \text{ p.u.}$$

$$(Z_{eq})_{s.c} = \frac{V_1}{I_1} \Big|_{V_2=0} = \frac{+8\cancel{\Omega_2}}{\cancel{+4\cancel{\Omega_2}}} = \frac{8}{4} \parallel \dots$$

$$(Z_{eff})_2 = \frac{V_2}{I_2} \quad (1)$$

$$\frac{W.K.T}{T_1} = CV_2 - DT_2$$

$$(Z_{\theta})_{2,0.C} = \frac{V_2}{I_2} \Big|_{I_1=0} \Rightarrow \begin{aligned} CV_2 - DI_2 &= 0 \\ CV_2 &= DI_2 \end{aligned}$$

$$\frac{V_2}{I_2} = \frac{D}{C} \parallel r_o$$

$$= \frac{D}{C}$$

$$(Z_{cr})_{2s.c} = \frac{V_2}{I_2} \Big|_{V_1=0}$$

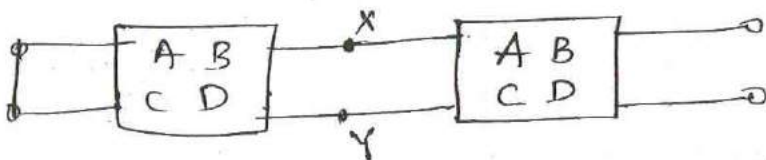
$$\begin{aligned} V_1 &= AV_2 - BT_2 \\ 0 &= AV_2 - BT_2 \end{aligned}$$

$$\frac{V_2}{I_2} = \frac{B}{A} \text{ y...}$$

$$= \frac{B}{A} \dots$$

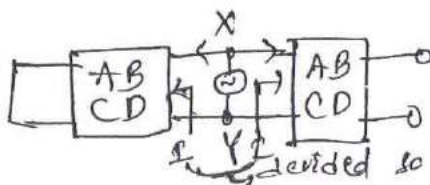
✱

(ph) Find Equi Impedance w.r. to x_2 .



$z_{xy} = z' = \frac{B}{A} \rightarrow$ corresponds to i/p port-res z

$$z_{xy} = z' = \frac{B}{A} \rightarrow \text{corresponds to o/p "1" e.o.c}$$

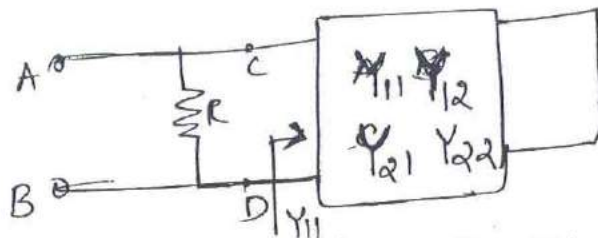


$\frac{1}{s} \parallel Y \parallel \frac{1}{s}$ decided so parallel for parallel ckt

$$\frac{z' z''}{z' + z''}$$

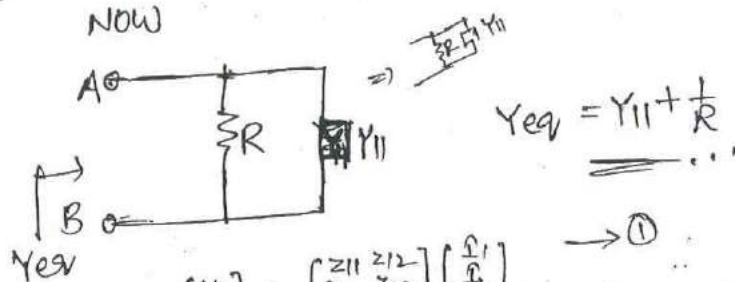
$$Z_{eq} = \frac{Z' * Z''}{Z' + Z''}$$

(pb) find eqn admittance w.r. to A & B.



sol:- w.r. to C & D eqn admittance = Y_{11}

Now



$$Y_{eq} = Y_{11} + \frac{1}{R}$$

Note: 1) $\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \rightarrow \textcircled{1}$

2) $\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \frac{1}{Z_{11}Z_{22} - Z_{12}Z_{21}} \begin{bmatrix} Z_{22} & -Z_{21} \\ -Z_{12} & Z_{11} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \rightarrow \textcircled{2}$

3) $\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \rightarrow \textcircled{3}$

4) $\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \frac{1}{Y_{11}Y_{22} - Y_{12}Y_{21}} \begin{bmatrix} Y_{22} & -Y_{21} \\ -Y_{12} & Y_{11} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \rightarrow \textcircled{4}$

from eqn 2 & 3

$$Y_{11} = \frac{Z_{22}}{\Delta Z} ; Y_{12} = \frac{-Z_{12}}{\Delta Z} ; Y_{21} = \frac{-Z_{21}}{\Delta Z} ; Y_{22} = \frac{Z_{11}}{\Delta Z}$$

where $\Delta Z = Z_{11}Z_{22} - Z_{12}Z_{21}$

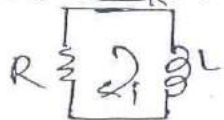
from eqn 1 & 4

$$Z_{11} = \frac{Y_{22}}{\Delta Y} ; -\frac{Y_{12}}{\Delta Y} = Z_{12} ; Z_{21} = \frac{-Y_{21}}{\Delta Y} ; Z_{22} = \frac{Y_{11}}{\Delta Y}$$

where $\Delta Y = Y_{11}Y_{22} - Y_{12}Y_{21}$

Note: RL CRT $\rightarrow T = \frac{L}{R}$
RC CRT $\rightarrow T = RC$

Why $(t \propto \frac{1}{R})$:-



Source free RL ckt

Assum $L \rightarrow$ charged inductor

so current is circulated $\Rightarrow$ dissipation in the resistor. $(P = i^2 R)$

at one instant Total energy in Inductor = 0 J at $t = 12 \mu\text{sec}$ let us assume.

$$R_1 = 10 \Omega ; P_1 = i^2 R_1 ; t = 12 \mu\text{sec}$$

$$R_2 = 20 \Omega ; P_2 = i^2 R_2 ; t = 10 \mu\text{sec}$$

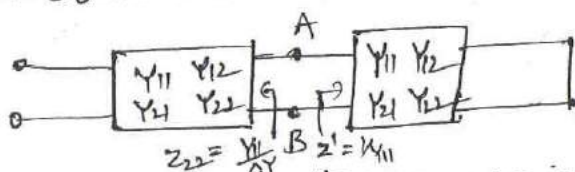
more P_{diss} so that in short time the energy will become zero.

$$R_3 = 100 \Omega ; P_3 = i^2 R_3 ; t = 3 \mu\text{sec}$$

As $R \uparrow \Rightarrow t \downarrow$ so inversely proportional.

Note :- Inductor acts as a current source.
Capacitor behaves as a voltage source.

(pb) Find equivalent impedance w.r. to A & B.



corresponds to ΔY N/W: A & B are o/p terminals.
 $\Delta Y = \text{open ckt}$

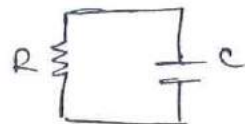
so Z_{aa} can be determined.

$$\text{but w.r.t } Z_{aa} = \frac{Y_{11}}{\Delta Y}$$

corresponding to N/W :-

o/p is short ckt
 $Z' = \frac{1}{Y_{11}}$

$Z' \rightarrow$ SC i/p impedance
 $Z_{22} \rightarrow$ OC o/p impedance.



$$R_1 = 10 \Omega ; P = \frac{V^2}{R_1} ; t = 10 \mu\text{sec}$$

time taken to disappear the energy.

$$R_2 = 20 \Omega ; P = \frac{V^2}{R} ; P_{\text{diss}} \downarrow$$

As $R \uparrow \Rightarrow P_{\text{diss}} \downarrow \Rightarrow$ energy loss $\downarrow \Rightarrow$ time taken to lose energy is high.

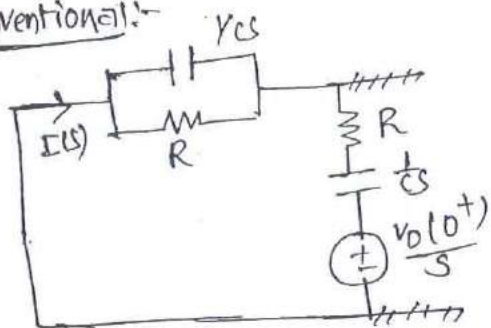
so $(t \propto R)$.

$$\therefore Z_{eq} = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

$$Z_{eq} = \frac{Z_{22} \cdot Z'}{Z_{22} + Z'}$$

Conventional:-

Q:1



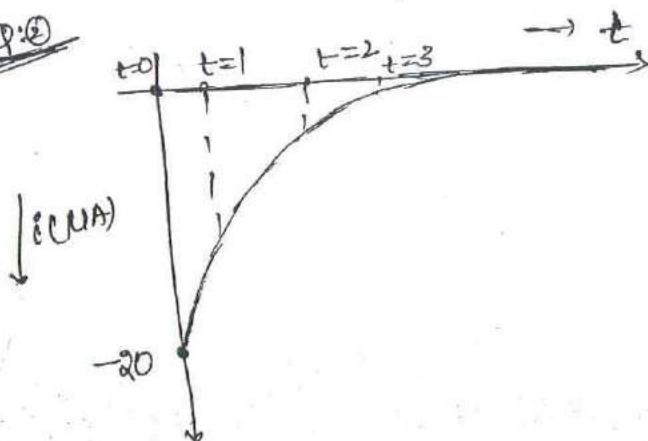
Generally all conventional pblm we go for laplace approach.

$$I(s) = \frac{-V_0(0^+)}{\frac{-s}{s}} \cdot \frac{R + \frac{1}{Cs}}{R + \frac{1}{Cs} + \frac{R \cdot (\frac{1}{Cs})}{R + \frac{1}{Cs}}}$$

$$\Rightarrow i(t) = \mathcal{L}^{-1}(I(s))$$

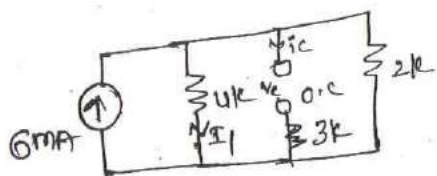
$$i(t) = -[14.46 e^{-2.62t} + 5.54 e^{-0.34t}] \text{ mA}$$

Q:2



Q:3

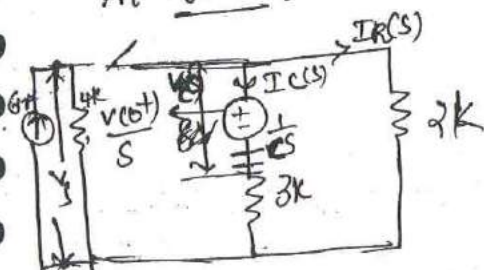
At $t=0^-$:



$$I_1 = \frac{6 \cdot (2)R}{8R} = 2 \text{ mA}$$

$$V_c(0^-) = (2 \text{ mA})(4 \text{ k}) = 8 \text{ V}$$

At $t > 0$:



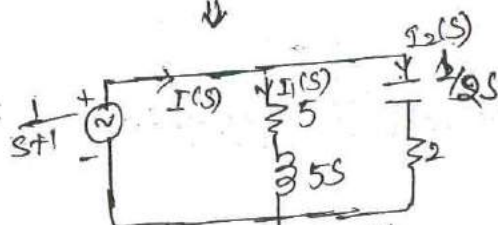
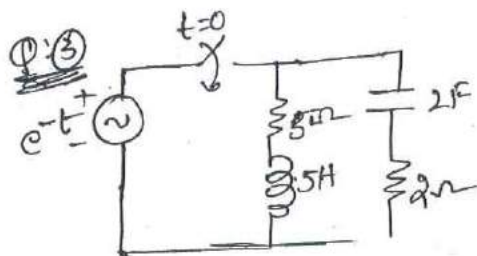
$$V_s = (4 \text{ k})(6 \text{ mA}) = 24 \text{ V}$$

$$I_R(s) = \frac{\frac{V(0^+)}{s}}{2 \text{ k} + 3 \text{ k} + \frac{1}{Cs}}$$

$$i_R(t) = \mathcal{L}^{-1}(I_R(s)) = (1.6 e^{-100t}) \text{ mA}$$

$$I_c = -I_R = -(1.6 e^{-100t}) \text{ mA}$$

Q:4



$$I(s) = I_1(s) + I_2(s)$$

$$I(s) = \frac{\frac{1}{s+1}}{5+5s} + \frac{\frac{1}{s+1}}{2+\frac{1}{2s}}$$

$$\Rightarrow i(t) = \mathcal{L}^{-1}(I(s))$$

$$= \left(\frac{2}{3} e^{-t} + \frac{1}{5} t e^{-t} - \frac{1}{6} e^{-t} \right) \text{ u(t)}$$

$$V_c = \frac{1}{C} \int_{-\infty}^t i_c(t) dt$$

Notes:

for including initial values take the limits as $-\infty$ to t .

At initial value $= 0 \Rightarrow$ then limits are 0 to t .

$$V_c = \frac{1}{C} \int_{-\infty}^0 i_c(t) dt + \frac{1}{C} \int_0^t i_c(t) dt$$

$$= 8 + \frac{1}{C} \int_0^t (-1.6 e^{-100t}) dt$$

$$= 8 + \frac{1}{2 \mu F} \cdot (-1.6) \left(\frac{e^{-100t}}{-100} \right)_0^t$$

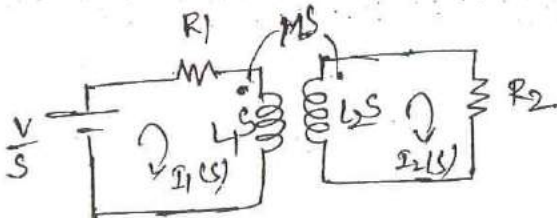
$$= 8 + \frac{1.6}{2 \times 10^{-4}} [e^{-100t} - 1]$$

$$= 8 + \frac{1.6}{2 \times 10^{-4}} e^{-100t} - \frac{1.6}{2 \times 10^{-4}}$$

.....

Q: 5

Sol:-



For Loop (1):-

$$-\frac{V}{S} + I_1(S)R_1 + L_1 S \frac{dI_1(S)}{dt} - MS \frac{dI_2(S)}{dt} = 0$$

$$I_1(S) [R_1 + L_1 S] - MS I_2(S) = \frac{V}{S} \rightarrow (1)$$

for Loop (2):-

$$I_2(S) [R_2 + L_2 S] - MS I_1(S) = 0 \rightarrow (2)$$

$$\text{from (2)} \quad I_2(S) = \frac{MS I_1(S)}{R_2 + L_2 S}$$

from Eq (1) $\Rightarrow$

$$I_1(S) [R_1 + L_1 S] - MS \left(\frac{MS I_1(S)}{R_2 + L_2 S} \right) = \frac{V}{S}$$

$$I_1(S) = \frac{(V/S)}{(R_1 + L_1 S) - \frac{(MS)^2}{(R_2 + L_2 S)}}$$

$$I_1(t) = \mathcal{L}^{-1}(I_1(S))$$

$$= [5 - e^{-t/5}] u(t)$$

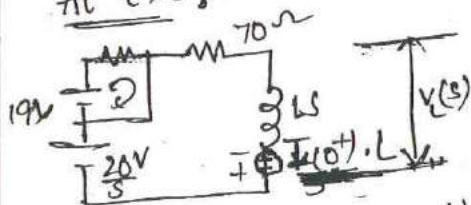
.....

Q: 6

At $t=0^-$:

$$i_L(0^-) = \frac{39}{130} = i_L(0^+)$$

At $t > 0$:



$$I(S) = \frac{(20/S) + L i(0^+)}{70 + L S}$$

$$\Rightarrow I(S) = \frac{20/S + (\frac{1}{3}) (\frac{39}{130})}{70 + S/3}$$

$$\Rightarrow i(t) = \left[\frac{2}{7} + \frac{1}{70} e^{-210t} \right]$$

.....

$$V_L = L \frac{di}{dt} \left(\frac{1}{3} \right) \frac{d}{dt} \left(\frac{2}{7} + \frac{1}{70} e^{-210t} \right)$$

$$= \frac{1}{3} \left(\frac{1}{70} \right) (-210) e^{-210t}$$

$$V_L(t=0) = \frac{1}{210} (-210) e^0 = -1V$$

Q: 8

Ans: $i(t) = \frac{E}{R_1} e^{-\left(\frac{R_1+R_2}{R_1 R_2}\right)t}$

$$V_0(t) = \frac{E R_2}{R_1 + R_2} \left[1 - e^{-\left(\frac{R_1+R_2}{R_1 R_2}\right)t} \right]$$

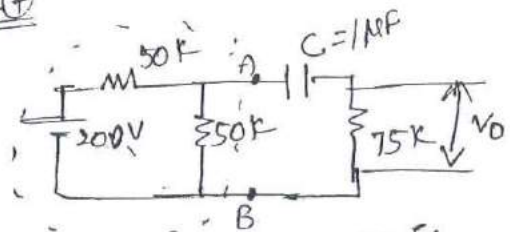
Q: 9

Ans: $V_C(t) = 12 - (t+2) e^{-t}$

Q: 10

Ans: $V_C(t) = 12 e^{-20t} (1 - 20t)$

Q: 7



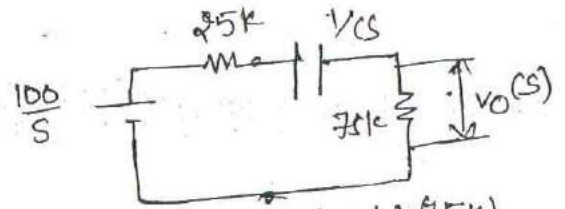
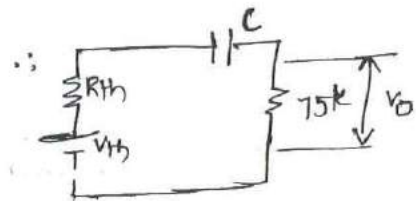
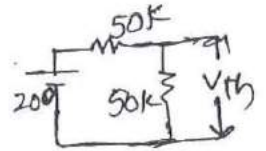
Thevenin's equivalent circuit is

case (i):

$$V_{th} = 100V$$

case (ii):

$$R_{th} = 25k\Omega$$



$$V_0(s) = \frac{(100/s)(75k)}{75k + 25k + \frac{1}{s}}$$

$$V_0(t) = \mathcal{L}^{-1}(V_0(s)) = 75 e^{-10t}$$

Note: Without Thevenin's theorem also we can do it, (by using Nodal analysis (or) Mesh analysis).

$$\therefore V_0(t) = 75 e^{-10(t-t_0)}$$

since the given circuit operated at $t = t_0$,

case (i): at $t = t_0 \Rightarrow V_0(t_0) = 75 e^{(0)} = 75V$

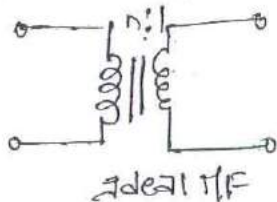
case (ii) at $t = 25ms$

$$\Rightarrow V_0(t) = 75 e^{-10(25 \times 10^{-3} - t_0)}$$

case (iii): at $t = \infty$

$$\Rightarrow V_0(t) = 75 e^{-\infty} = 0V$$

(Pb) Find Z, Y & ABCD parameters of the N/w shown.



Sol:- $\frac{V_1}{V_2} = \frac{n}{1}$
 $V_1 = nV_2 \rightarrow (1)$

$\frac{I_2}{I_1} = \frac{n}{1}$
 $I_2 = nI_1 \rightarrow (2)$

Note:-

Self & mutual inductance are ∞ for ideal T/F.

from (1) & (2) we can't find out the Z & Y parameters. Since (V_1, V_2) interms of (I_1, I_2) for Z, & (I_1, I_2) interms of (V_1, V_2) for Y parameters.

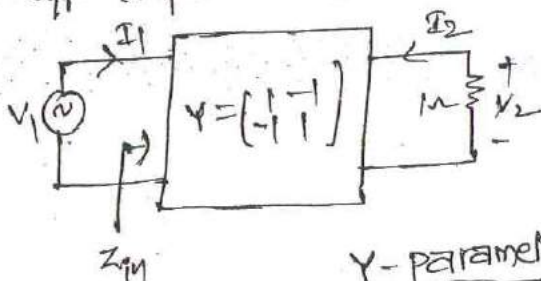
Note:- For ideal T/F it's not possible to find impedance & admittance values, since self & mutual inductance of ideal T/F are infinite.

$A = \frac{V_1}{V_2} = n$; $D = \frac{-I_1}{I_2} = \frac{1}{n}$

(since I_2 actually opposite to two port N/w current direction)

(Pb)

Find d/p impedance of the ckt shown?



Sol:-

$V_2 = -1 \times I_2$
 $\Rightarrow I_2 = -V_2$

Y-parameter eqns are:-

$I_1 = Y_{11}V_1 + Y_{12}V_2 \Rightarrow I_1 = 1V_1 - V_2$
 $I_2 = Y_{21}V_1 + Y_{22}V_2 \Rightarrow I_2 = -V_1 + V_2$

but $I_2 = -V_2$

$-V_2 = -V_1 + V_2$
 $\Rightarrow +V_1 = +2V_2$

$V_1 = 2V_2 \rightarrow (1)$

$Z_{in} = \frac{V_1}{I_1}$

$I_1 = 1V_1 - (\frac{V_1}{2})$

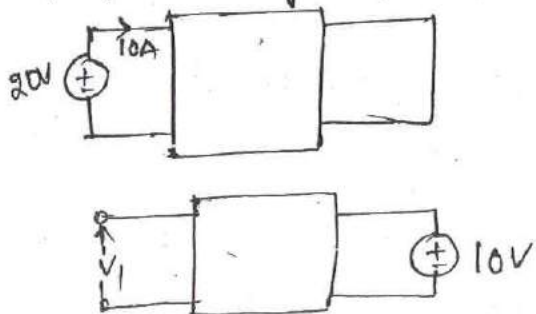
$I_1 = \frac{V_1}{2}$

$\Rightarrow \frac{V_1}{I_1} = 2 = Z_{in}$

$\therefore I_2 = -2V_2 + V_2 = -V_2$

(Pb)

Find V_1 of the figure shown.



This can be solved by using theorem (or) two port N/w parameters.

Here op is short ckt & another one: open ckt. so we can't write Z_{in} parameters, so we can go for h-parameters.

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

from the given data

$$V_2 = 0 \Rightarrow h_{21} = \frac{I_2}{I_1} = \frac{-2}{10} = -\frac{1}{5} \dots$$

$$\text{when } I_1 = 0 \Rightarrow h_{12} = \frac{V_1}{V_2}$$

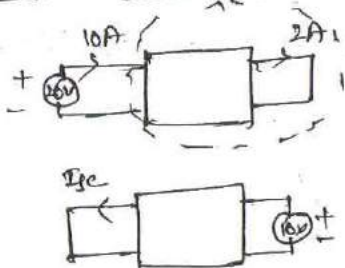
$$= \frac{V_1}{10} \dots$$

Note:- When ever in the pblm if they doesn't mention about the ckt by default take it as Reciprocity.

$$\text{so } h_{12} = -h_{21}$$

$$\Rightarrow -(-\frac{1}{5}) = +\frac{1}{10} \Rightarrow V_1 = 2V$$

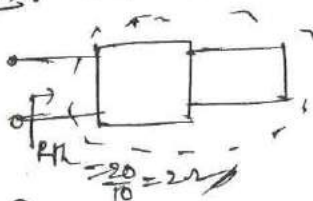
Note:- By using Theorems:-



To verify the reciprocity theorem, make the o.c or s.c. so $I_{sc} = 1A$

$$\frac{P_{sc}}{P_{oc}} = \frac{2}{20} = \frac{1}{10} \Rightarrow I_{sc} = 1A$$

Now R_{th} : while finding the R_{th} make all sources are deactivated.



physical connections are same $\Rightarrow$ ckt elements are same.

$$\text{for fig 1: } R_{th} = \frac{20}{10} = 2\Omega$$

from original NW, the port 1 is o.c so we can write

$$V_{oc} = I_{sc} \cdot R_{th} = (1A)(2\Omega) = 2V = V_1$$

(P6)

for the ckt shown if $B = \frac{2s+1}{s^2}$

find the value of $z(s)$,

$$\text{sol:- } V_1 = AV_2 - BI_2$$

$$B = -\frac{V_1}{I_2} \mid V_2 = 0$$

with respect to i/p terminals

$$Z_{sc} = \frac{V_1}{I_1} = \frac{1}{s} + \frac{1(z(s))}{1+z(s)}$$

$$V_1 = I_1 \left[\frac{1}{s} + \frac{z(s)}{1+z(s)} \right] \rightarrow (1)$$

$$= \begin{bmatrix} 1 & \frac{1}{s} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & z(s) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{s} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1+z(s) & z(s) \\ 1 & 1+z(s) \end{bmatrix} = \begin{bmatrix} 1+z(s) & z(s) + \frac{1+z(s)}{s} \\ 1 & 1+z(s) \end{bmatrix}$$

$$z(s) + (1 + z(s)) = \frac{2s+1}{s}$$

or

$$I_2 = -I_1 \cdot \frac{1}{1+z(s)} \rightarrow (2)$$

$$(s z(s) + 1 + z(s)) = 2s + 1$$

$$\text{Compare } z(s) \left(\frac{1}{s} + 1 \right) = \frac{2s+1}{s} - 1$$

$$z(s) = 2$$

$$\Rightarrow z(s) \left[1 + \frac{1}{s} \right] = \frac{(s+1)}{s} \Rightarrow z(s) = 1$$

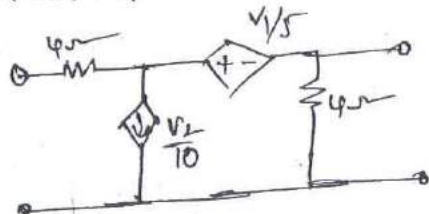
$$\therefore B = -\frac{V_1}{I_2}$$

from (1) & (2)

$$\Rightarrow B = \frac{(1)}{(2)} = \frac{2s+1}{s^2}$$

$$\Rightarrow z(s) = \frac{1}{s}$$

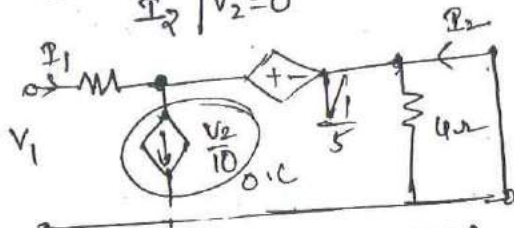
(Pb)



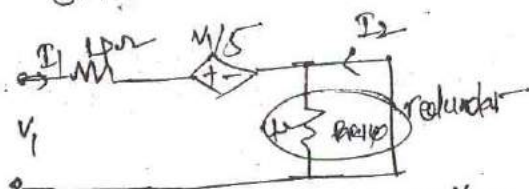
find B & D?

sol:-

$$B = \frac{-V_1}{I_2} \mid V_2 = 0$$



Since $V_2 = 0 \Rightarrow \frac{V_2}{10} \text{ branch} = 0$



$$I_2 = -I_1 \rightarrow (1)$$

$$V_1 = 4I_1 + \frac{V_1}{5}$$

$$V_1 \left[1 - \frac{1}{5} \right] = 4(-I_2)$$

$$V_1 \left[\frac{4}{5} \right] = 4(-I_2)$$

$$\therefore \frac{V_1}{I_2} = -5 \Rightarrow B = 5 \dots$$

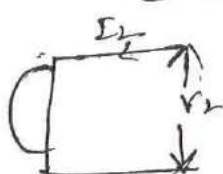
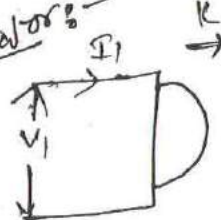
$$D = \frac{-I_1}{I_2} \mid V_2 = 0$$

$$= \frac{+I_1}{-I_2}$$

$$D = \frac{+1}{-1} \dots$$

(Pb)

Gyration



units for $k = \Omega$

sol: Note:-

- 1) In the designing of Active filters we will use the concept of Gyrator.
- 2) Gyrator is a passive, linear, loss-less and Non-reciprocity two port N/W.

$$V_1 = -k I_2 \rightarrow (1)$$

$$V_2 = k I_1 \rightarrow (2)$$

$$[Z] = \begin{bmatrix} 0 & -k \\ k & 0 \end{bmatrix}$$

$$V_1 = Z_{11} I_1 + Z_{12} I_2 \rightarrow (3)$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2 \rightarrow (4)$$

from (1) & (3) & (2) & (4)

$$Z_{11} = 0; Z_{12} = -k$$

$$Z_{21} = k; Z_{22} = 0.$$

* Gyrator relates voltage at one port to current of the other port.

* Gyrator is a passive, linear, lossless & non-reciprocity two-port network.

* Gyrator is used in the designing of Active filters.

Note: ** when Current is present at head of the arrow mark -ve sign is present.

** when Current is present at tail of the arrow mark +ve sign is assign.

from eq (1) $\Rightarrow I_2 = -\frac{V_1}{K} \rightarrow (4)$

eq (2) $\Rightarrow I_1 = \frac{V_2}{K} \rightarrow (5)$

$$I_1 = Y_{11} V_1 + Y_{12} V_2 \rightarrow (6)$$

$$I_2 = Y_{21} V_1 + Y_{22} V_2 \rightarrow (7)$$

By comparing (4) & (6) & (5) & (7) we get

$$Y_{12} = \frac{1}{K} \quad Y_{21} = -\frac{1}{K}$$

$$Y_{11} = 0 \quad Y_{22} = 0$$

$$\therefore [Y] = \begin{bmatrix} 0 & \frac{1}{K} \\ -\frac{1}{K} & 0 \end{bmatrix}$$

Now $V_1 = AV_2 - BI_2 \rightarrow (8)$ from eq (1) & (8)

$I_1 = CV_2 - DI_2 \rightarrow (9)$ eq (2) & (9)

$$A=0; B=K$$

$$C=\frac{1}{K}; D=0$$

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 0 & K \\ \frac{1}{K} & 0 \end{bmatrix}$$

Note:- corresponds to Gyrator $\rightarrow$ lossless N/w but units are Ω } how we will see?

let us total power consumed is P_T :

$$P_T = V_1 I_1 + V_2 I_2$$

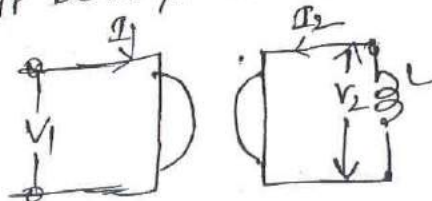
$$P_T = ((-K I_2) I_1) + ((K I_1) I_2)$$

$\boxed{P_T = 0}$ from this we can conclude
as a lossless N/w

Note:-

If a/p behaves as capacitor $\Rightarrow$ o/p behaves as Inductor & vice versa

$$V_2 = -L \frac{dI_1}{dt}$$



$$V_2 = -L \frac{di_2}{dt}$$

$$KI_1 = -L \frac{d}{dt} (-V_1/K)$$

$$KI_1 = \frac{L}{K} \frac{dV_1}{dt}$$

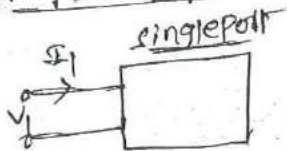
$$I_1 = \frac{L}{K^2} \frac{d(V_1)}{dt}$$

but w.r.t $\Rightarrow C = \frac{L}{K^2}$

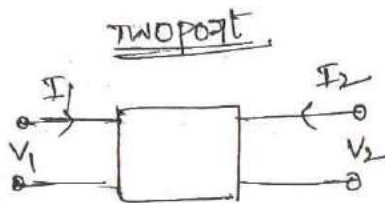
$$I_C = C \frac{dV}{dt}$$

- * In the Gyration at output is connected to the inductor input behaves as a capacitor.
- * at output is connected to capacitor i/p behaves as a inductive.

Network functions:-



Amittance $Z(s) = \frac{V(s)}{I(s)}$
 ↓
 This is the common name for both functions.



1. Driving point impedance

$$Z_{11}(s) = \frac{V_1(s)}{I_1(s)}$$

$$Z_{22}(s) = \frac{V_2(s)}{I_2(s)}$$

2. Driving point admittance

$$Y_{11}(s) = \frac{I_1(s)}{V_1(s)}$$

$$Y_{22}(s) = \frac{I_2(s)}{V_2(s)}$$

3. Transfer impedance

$$Z_{12}(s) = \frac{V_1(s)}{I_2(s)}$$

$$Z_{21}(s) = \frac{V_2(s)}{I_1(s)}$$

4. Transfer admittance

$$Y_{12}(s) = \frac{I_1(s)}{V_2(s)}$$

$$Y_{21}(s) = \frac{I_2(s)}{V_1(s)}$$

5. Transfer voltage ratio

$$G_{12}(s) = \frac{V_1(s)}{V_2(s)}$$

$$G_{21}(s) = \frac{V_2(s)}{V_1(s)}$$

6. Transfer current ratio

$$\alpha_{12} = \frac{I_1(s)}{I_2(s)}$$

$$\alpha_{21} = \frac{I_2(s)}{I_1(s)}$$

Note:- Difference b/w N/W parameters & N/W funn :-

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} \text{ o.c}$$

$$Y_{11} = \frac{I_1}{V_1} \bigg|_{V_2=0} \text{ s.c}$$

Always N/W parameters are calculated based on some predefined funn i.e o.c & s.c. But in this N/W funn no need to go for predefined funn.

2. Suppose we want to design one N/W i.e $(Z_{11}, Z_{12}, Z_{21}, Z_{22})$ all are must be known. but in N/W functions only one single value is required.

$$\begin{matrix} Z_a & Z_b \\ \hline Z_c & \end{matrix} \text{ (TN/W)}$$

* N/w parameter is defined at predefined condition (either o.c. or s.c.).

$$Z_{11} = \frac{V_1}{I_1} \bigg|_{I_2=0} \text{ o.c.}$$

$$Y_{11} = \frac{I_1}{V_1} \bigg|_{V_2=0} \text{ s.c.}$$

** TO Find N/w function NO predefined condition is required.

** TO represent the N/w simultaneously four parameters are required.

TO represent T N/w $Z_{11}, Z_{12}, Z_{21}, Z_{22}$ all the parameters are required.

** By using only one N/w function it is possible to represent the N/w.